

One corpus, *three* sources.

This morning Daniel set the app up for his son for the first time. The son is going for all of Mishnayos by his bar mitzvah. By the evening it was obvious the app believed something about his life that was not true — so we spent the evening designing the part that makes it true.

6	63	~4,192	3	1
SEDARIM	MASECHTOS	MISHNAYOS	SOURCES FEEDING ONE GOAL	DEADLINE

WHERE IT STARTED

The app assumed a straight line. His life is a confluence.

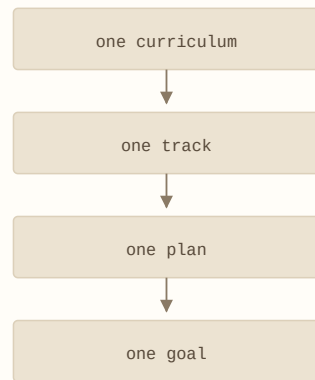
THE BIG ONE: app assumes one curriculum -> one track -> one plan -> one goal. Reality is one corpus (all Mishnayos) fed by three sources: Home (self-study, goal-bearing, scheduled), Rebbe (own masechta, no goal, as-and-when), School (own set, no goal, as-and-when). All accumulate to the same completion.

DANIEL — THE SESSION'S FIRST SUBSTANTIVE ENTRY

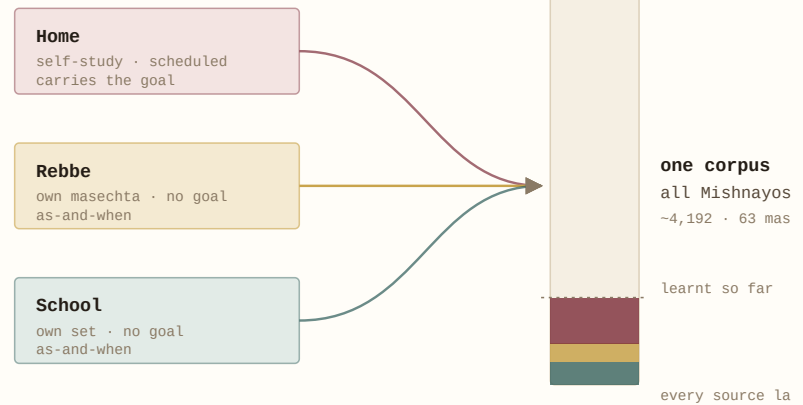
Everything that followed is a consequence of that one sentence. The corpus is shared; the goal is single; the ways ground gets covered are three, and two of them nobody at home controls. The coach's first useful move was to say what rebbe and school are *not*: they are not tracks, because a track

implies a plan. They are **attribution sources** posting completions against one shared corpus, and exactly one track carries the goal and does the forecasting.

What the app assumed



What his life actually is



Three inflows, one vessel, one waterline. The design problem is not that there are three sources – it is that only one of them takes instructions.

HOW IT WENT

Not a technique. A conversation.

No card decks, no six hats, no timed divergence. Daniel dumped straight into content and the session ran as a live design dialogue for 161 logged entries — an idea, a consequence, a push back, a scope lock, and round again. The good parts are the places where somebody was wrong.

Daniel names the big one — and three “minor changes”

Grey box for partial completion; reordering must discard stale backlog; enter expected weekly output from the other programs. Two of the three turn out to be the big one in disguise.

The reframe: rebbe and school are not tracks

A track implies a plan. They are attribution sources posting completions against one shared corpus. Exactly one track carries the goal and does the forecasting.

The spine: the plan is derived, the events are the facts

Backlog and today's schedule are computed from corpus + sequence + completions + goal date, never stored. Reorder, missed days and school coverage collapse into one trigger with three causes: inputs changed, recompute.

● **The calendar problem, already solved four messages earlier**

Schools do not follow bein hazmanim; every country differs; every school differs even within a country. The coach circles inferred calendars and per-locale zmanim for three exchanges before noticing that Daniel's "40 weeks per year" had already made the model calendar-free.

○ **The double-count landmine**

Carving out Masechta Berachos because school is doing it, *and* subtracting 10/week for school, are two mechanisms for the same contribution. Do both naively and the home target comes out too low and he silently falls behind.

● **Three scope locks and a governing principle**

Yom tov and bein hazmanim exist only to make sub-track estimates honest. No sub-goals — the deadline is a property of the one goal-bearing track. No deadline, no required rate. And: keep it as simple as possible, applied retroactively to everything already generated.

● **The throwaway that reorganised the schema**

The lifetime learning record arrives almost as an aside. The coach's "completions are a set" dies on the spot, and with it the whole collision problem: school covering ground already learnt at home is not a conflict, it is chazara.

● **Shabbos stops being a scheduling detail**

He learns at home on shabbos, so the main track has no excluded days at all. But he cannot log. The coach calls it self-healing; Daniel refuses the shrug; the app gets a shabbos mode.

● **Sub-tracks harden**

Drag any masechta, perek or mishna in and it leaves the main track. Two types. One school sub-track per academic year, five other ongoing ones. Each track has exactly one current position.

● **The reversal that saved the codebase**

Two masechtos on the main track must be impossible. The existing planner, task generation and sequencing are untouched; everything new lives in sub-tracks.

Where the coach was wrong.

Six times the analyst's own proposal was withdrawn or overruled. Each one is in the log with its correction attached, because the reversals are where the design actually got made.

THE COACH PROPOSED

~~Completions are a set, not a list. Simplest possible model — one mishna is either done or not.~~

DANIEL **OVERRULED**

Lifetime learning record: every learning event kept globally — tap a mishna and see how many times he has learnt it and when, across his whole life. Plus a lifetime report.

WHAT STANDS

An append-only event log — mishna, date, source. It cost nothing extra: the per-source velocity report already needed timestamped, attributed events. One table yields goal progress, times-learnt, history, velocity and attribution.

THE COACH PROPOSED

~~Avoid inventing two sub-track types. Keep one form and make the exclusion line optional and collapsed.~~

DANIEL **OVERRULED**

There **are** two sub-track types. A school sub-track runs September to July and lasts exactly one year. Not an arbitrary window.

WHAT STANDS

Two types — and the type *is* the simplification. Because a school year is a real fixed recurring structure, typing it lets the app fill the dates, offer “add next year”, drop the exclusion fields entirely and refuse nonsense windows. Generic-and-flexible would have made Daniel do all of that by hand.

THE COACH PROPOSED

~~A sub-track with several masechtos gets one row per masechta, each with its own progress bar and its own next mishna pointer.~~

DANIEL **OVERRULED**

Each track has exactly **one current position**. A sub-track holding several masechtos needs an order, like the main track. Only one thing is being learnt at a time per track.

WHAT STANDS

Every track is an ordered sequence over assigned ground with a single current position. The home screen stays one row per track however much ground is assigned — and there is exactly one “next”, so the +1 button never has to ask which masechta.

THE COACH PROPOSED

~~One kind of track, one kind of row. Main track and sub-tracks render identically; the deadline is the only difference.~~

DANIEL **CORRECTED**

The personal main track keeps its **current** behaviour — it still generates tasks and scheduled items exactly as it does today.

WHAT STANDS

Withdrawn. A planned track produces **tasks**; an unplanned track produces a **position**. The presence of a plan, not the presence of a deadline, is what splits the UI. The home screen is mixed — and cheaper to build than the unified version.

THE COACH PROPOSED

~~Rate and active-weeks estimates belong in the main track's goal settings, not on the sub-track — a sub-track has no goal of its own, so the numbers are not its property.~~

DANIEL **OVERRULED**

The whole form is: free-text name, mishnayos per week, exclusion checkboxes, adjustable weeks per year. All of it at creation, in one place.

WHAT STANDS

Withdrawn. Everything about a source lives where the source is created. Model purity loses to user mental model.

THE COACH PROPOSED

~~Shabbos capture needs no special logic — with no streaks and a derived plan, an unticked day nudges tomorrow's target up and motzei's ticks drop it straight back. Self-healing.~~

DANIEL PUSHED BACK

If shabbos is ticked as a learning day, the inability to log on it must actually be **dealt with**, not shrugged off.

WHAT STANDS

Shabbos mode: capture locked from candle-lighting, a read-only view of what is planned, a motzei catch-up card, and events dated to the day they were learnt. It became the most distinctive feature in the whole design.

And once, Daniel overruled Daniel

Mid-session he confirmed that the main track could hold several masechtos at once, the same as a sub-track. Later he reversed it outright — two masechtos on the main track must be **impossible**. That single reversal deleted an open question the coach had just raised (if the main track holds two masechtos, which one does today's task list draw from, and in what proportion?) and left the existing planner completely untouched. The main track only ever sees its remaining pool shrink and its daily target adjust — both already handled by the recompute engine.

THE CENTRE OF THE WHOLE THING

One equation.

Somewhere in the middle of the session it became clear that every single thing being discussed was either an input to the numerator or an input to the denominator. Nothing else was actually being designed.

$$\text{daily target} = \frac{\text{remaining} - \sum \min(\text{rate} \times \text{weeks left in window}, \text{ground left})}{}$$

REMAINING

The corpus minus every event logged to date — which already includes everything school and rebbe have ticked. The past is counted as fact.

$\Sigma \text{ MIN}(\dots)$

Forward-only, summed over every sub-track. Ground assigned to a sub-track caps its contribution at that ground; a sub-track with no ground contributes pure quantity. One formula, two modes, no mode switch — and no way to count a source twice.

DAYS TO DEADLINE

He learns at home on shabbos, so the main track has no excluded days. Every calendar concept in this design lives in one place: the weeks-per-year count on a sub-track.

Why the estimates earn their place — illustrative, as worked in the session: ~4,192 mishnayos with three years to the bar mitzvah.

ignoring school & rebbe



~3.8 / day

counting them honestly



~2.2 / day

School at 10/wk $\times$ 39 weeks and rebbe at 5/wk $\times$ 44 weeks contribute roughly 1,830 mishnayos over the three years. The estimates roughly **halve** the apparent daily load. Without them the app demands nearly double what he actually needs — and a boy who is told to do 3.8 when 2.2 is the truth is being told, every day, that he is failing.

Run it backwards and you get the free version

Same machinery, both directions. Deadline set → solve for **rate**: how much per day. No deadline → solve for **date**: at this pace you finish in March 2029. And with no deadline you need no estimates at all, because actual observed velocity already contains school and rebbe. That is the build order: attribution plus measured velocity ships with **zero configuration**; the entire estimate apparatus unlocks only when a deadline exists.

The landmine it defuses

Reserving ground (“carve out Berachos, school is doing it”) and forecasting a rate (“subtract 10/week for school”) are two mechanisms for the same contribution. Do both naively and school pays in twice, the home target comes out too low, and he falls behind without the app ever saying so. The **min()** is what makes that impossible: with ground assigned, the rate stops adding contribution and starts predicting *when* the assigned ground completes.

The principles that survived the evening.

01

The plan is derived. The events are the only stored facts.

Backlog and today's schedule are computed from corpus + sequence + events + goal date, never stored. Reordering, missed days and school coverage are one trigger with three causes.

Learning out of order needs no special case at all — it simply shrinks the remaining set and lowers tomorrow's target.

02

Exactly one deadline, on the main track.

No sub-goals, ever. No “finish Seder Moed before Pesach”. A sub-track has no goal and no required rate — but it does have progress. The equation is solved once, globally.

03

The main track never changes shape.

One masechta at a time, one plan, one deadline, the sequencing and task generation exactly as they are today. Everything new lives in sub-tracks. The main track only sees its remaining pool shrink.

04

Calendars touch nothing but a sub-track's weeks.

Yom tov and bein hazmanim exist solely to make a sub-track's throughput estimate honest. No term imports, no per-country school logic, no dates in the model at runtime — one rate and one week count each.

05

Chazara, not collision.

School covering ground already learnt at home is not a conflict to resolve; it is a second learning event, and a good one. Goal progress counts **distinct** mishnayos; the lifetime record counts **all** events. Nothing to dedupe, no dialog, and nothing lost.

06

Build in the order that needs no configuration.

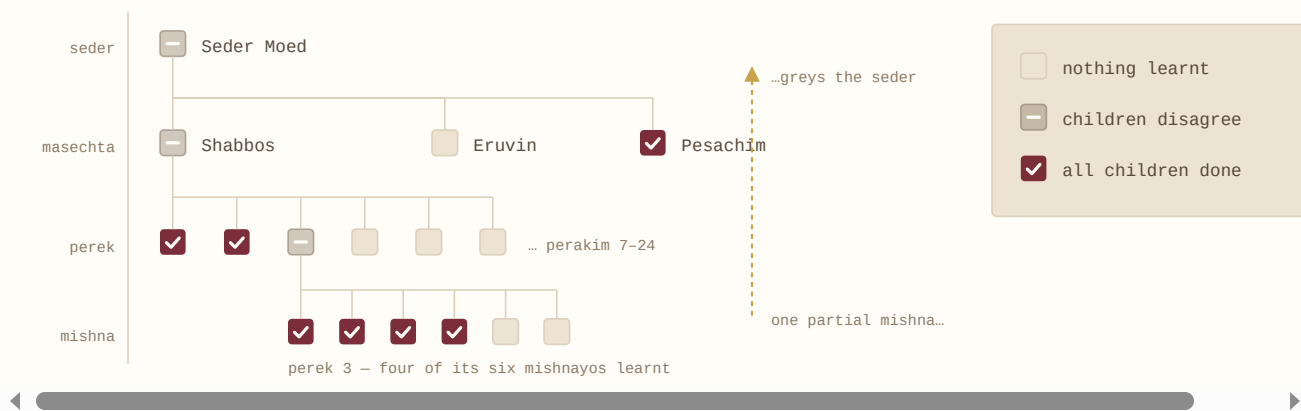
Attribution and measured velocity work with zero setup and already project a finish date. Estimates, active weeks and rates are progressively disclosed — hidden until a deadline exists, then shown exactly when they start to mean something.

Four things the app has to be able to draw.

Each of these started as a sentence in the conversation and only became real once it was drawn. They are the load-bearing pictures of the design.

1 • The grey box, cascading

Minor 1 in Daniel's list, and a prerequisite for everything else: completing one perek leaves the whole masechta reading unticked. He wants a grey box. The coach's contribution was to notice that partiality *cascades* — it is a rule for how any parent renders when its children disagree, at every level of the hierarchy, not a masechta-level fix.



Once school or rebbe cover part of a masechta, partial stops being an edge case and becomes the normal state of the tree. The same tri-state rendering is then reused in four places: the main corpus, the backfill browser, the drag-in-ground picker, and every sub-track's own progress view.

2 • The contribution timeline

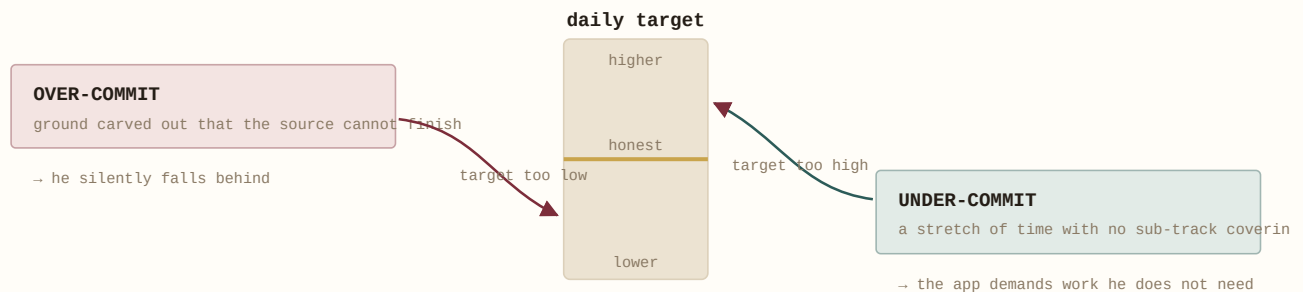
Once a sub-track can have a window, the forecast stops being a flat rate and becomes a timeline of overlapping contribution windows — materially more accurate over a three-year horizon. School years chain one after another; an ongoing track just runs. And the holes become visible.



A future sub-track is legal and useful: "starts Sep 2027, 12/week" contributes to the forecast before it exists in reality and before it has a single tick. An uncovered stretch silently forecasts zero help and inflates the daily target — so the app names it rather than letting it sit there.

3 · The two ways a forecast lies

Both failure modes are quiet, both move the daily target, and they move it in opposite directions. Both are cheap to detect from data the app already holds.



The rate validates the carve-out

If a sub-track is assigned more ground than its rate can finish before the deadline, the shortfall comes back to the main track with a warning: "School will not finish Masechta Shabbos by 12 Sivan - 40 mishnayos come back to you." Carving out ground a source never completes is the quiet way to miss the bar mitzvah.

Gap detection along the timeline

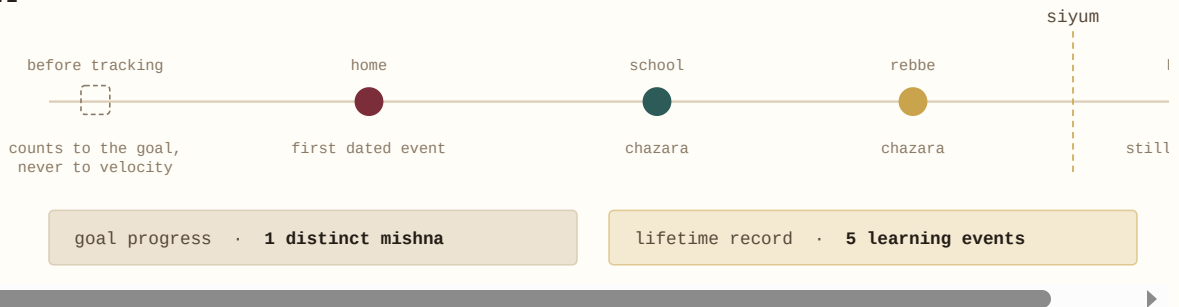
Any period between now and the deadline with no covering sub-track forecasts zero help. The app surfaces it as a question rather than an error — finished, or not yet planned? Same family as the warning on the left: both are holes in the forecast, and both are visible in data already held.

4 · One mishna, a whole life

The idea that arrived last and reorganised the storage model. Completions are not a set; they are an append-only event log of mishna + date + source. Goal progress counts **distinct** mishnayos. The lifetime record counts **all** events. A second learning never advances the bar mitzvah goal and always enriches the record.

Berachos 1:1

illustrative



Because the log carries dates, the corpus can also show what is warm and what is cold – a chazara prompt driven by the same data, with no new storage. Flagged optional against the simplicity mandate. And because the record is a lifetime record, the app outlives the goal: the bar mitzvah is one milestone in a record that keeps going.

THE SCREENS THAT WERE AGREED

Three surfaces, and nothing else new.

The sub-track forms went through five revisions in the log and ended smaller each time. What is drawn here is the last version: every number visible, every number editable, and nothing hidden inside an average that needs explaining.

NEW SUB-TRACK

SCHOOL YEAR

NAME

School

ACADEMIC YEAR

Sep 2026 – Jul 2027 already added

Sep 2027 – Jul 2028

NEW SUB-TRACK

ONGOING

NAME

Rebbe

MISHNAYOS PER WEEK

5

NOT LEARNING DURING

Bein hazmanim

Sep 2028 – Jul 2029

Only years between today and the deadline are offered. One per year – the impossible choice is simply not shown.

MISHNAYOS PER WEEK

12

WEEKS PER YEAR

39

Pre-filled, always overridable.

Add

Add next year

No exclusion checkboxes. A school-year figure already has the holidays inside it; applying exclusions on top would discount the breaks twice and understate what school contributes. “Add next year” clones this form and shifts the window, so programming four years costs one tap each. No year numbering – numbering differs by country, so a school sub-track is identified by its academic period.

TODAY

MON 24 AUG

TODAY · MAIN TRACK

2.2 / DAY

☐ Shabbos 7:2

☐ Shabbos 7:3

☒ Shabbos 7:1

SUB-TRACKS

3

Rebbe

Beitzah 2:6

+1

up to...

A convenience that pre-fills the number below – never a tick-box that traps an estimate you disagree with.

WEEKS PER YEAR

44

END DATE (optional)

none

Add

Yom tov was deleted as a checkbox late in the session – one exclusion survives, and it is bein hazmanim. Up to five ongoing sub-tracks; one school sub-track per academic year from now to the goal. Assigning ground is **phase two** and entirely optional: drag any masechta, perek or individual mishna into a sub-track, whenever the school or rebbe announces what they are learning, and it disappears from the main track.

Every open front, in one place

Daniel asked for easy capture of what he learnt with his rebbe or at school **three separate times** across the session. It is clearly the thing that decides whether the app gets used at all, so it got the top of the screen and no navigation.

The screen is deliberately **mixed**, not unified. Today's tasks render exactly as they do now, because the main track has a plan and a plan produces tasks. Beneath them sit the sub-tracks, which have no plan and therefore produce a **position** — one row each, however many

School

Berachos 4:1

+1

up to...

Summer shiur

Megillah 1:4

+1

up to...

masechtos are assigned, because every track has exactly one current position.

- **+1** for the single daily mishna.
- **Up to...** for everything else. People report learning by endpoint, not by count — “we finished perek 3” — so tap where he got up to and everything before it fills in. One gesture for any amount.
- **Tick-to-here also fixes this morning.** Backfilling hundreds of already-learnt mishnayos one tap at a time becomes a single “up to here” per masechta.
- **The source is free.** It is implied by which row he ticked in — no dropdown, no question, no extra taps — and that one column is what unlocks every report in the design.

SHABBOS

It came in as a scheduling exclusion. It left as the whole point.

It entered the session as a checkbox — exclude shabbos and yom tov from the schedule. Then Daniel answered the open question about the denominator: **yes, he learns at home on shabbos**. Which removes the exclusion entirely — the main track has no excluded days, and the denominator is simply every day from now to the deadline — and replaces it with a much better problem. He learns, and he cannot log.

The coach's first answer was that this needs no special handling: with no streaks and a derived plan, an unticked day nudges tomorrow's target up a fraction and motzei's ticks drop it straight back. Self-healing. Daniel did not accept it.

*If shabbos is ticked as a learning day, the inability to log on it must actually be **dealt with**, not shrugged off.*

DANIEL

And then, when the coach offered the alternate reading — should the app *refuse* to be used on shabbos rather than merely tolerate the gap? — he took it further than the coach had:

*The app must be **not clickable** on shabbos, and by extension yom tov. Capture is locked, no writing. After shabbos he is expected to tick what he learnt.*

DANIEL — SHABBOS MODE CONFIRMED

capture open

lock early

capture locked

unlock

candle-lighting

tzeis

Erev shabbos

a false lock is harmless – a false unlock is the failure that matters

Motzei shat

CAPTURE LOCKED – read only
the plan is visible, nothing can be written

kosher_dart is already a dependency, and one setting held once – location, and Israel or chutz la'aretz – drives the whole window. flutter_local_notifications is already there too, so the motzei reminder fires off the same zmanim and lands exactly when the catch-up card becomes tappable.

SHABBOS

LOCKED

PLANNED FOR SHABBOS

14 MISHNAYOS

Shabbos 7:2 – 7:4

Shabbos 8:1 – 8:7

Shabbos 9:1 – 9:4

On target. This reading will not change until motzei shabbos.

nothing on this screen can be tapped

MOTZEI SHABBOS

CATCH-UP

SHABBOS · 14 MISHNAYOS PLANNED

Learnt them all?

Yes, all of it

Adjust...

Ticks are dated to shabbos, not to now.

Four small pieces

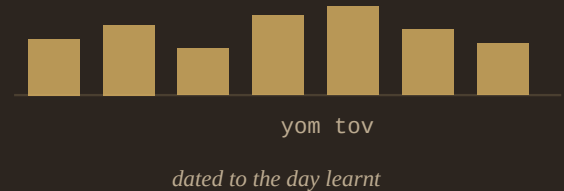
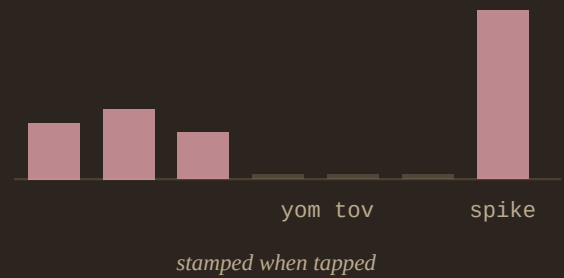
Show the target before it comes in. The erev view says what there is to learn.

Why the dating is not cosmetic

Three days of yom tov learning, all stamped with the moment they were tapped:

- **Freeze the on/off-target indicator** for the duration. Never render “behind” on a day he cannot log.
- **A motzei catch-up card.** One tap to confirm the whole lot, or adjust. It generalises to yom tov — up to three consecutive un-loggable days — with no extra design.
- **Date the events to the day learnt,** not to the moment they were tapped.

The locked screen and the “show the target before shabbos” requirement turned out to be the **same screen**: a read-only view of what is planned, with no buttons on it. One thing to build, and the most useful possible thing to show on a locked screen. The app cannot stop a phone being used on shabbos — but it can refuse to record, which is the part it is responsible for.



The left-hand version reports a fake spike and zero yom tov learning. The lifetime record would be telling a lie about his life — and the velocity figure that decides whether he is on target would be built on it.

The same dating rule catches a bug that was already live: Daniel backfilled hundreds of already-learnt mishnayos on setup day. Stamped with the tap date, the velocity report opens with a several-hundred-mishnayos spike on day one and a permanently distorted baseline. So there are three date states — **dated**, **motzei-backdated**, and **learnt before tracking started**. Backfill is an opening balance: it counts in the balance, never as this month's income.

THE DELIVERABLE

What actually gets built, in order.

The session was asked for a change list: minors first, then the big one. It ends with the minors turning out to be most of the big one, and the big one turning out to touch the main track not at all.

STAGE 1

The minors

STAGE 2 — THE BIG ONE

Sub-tracks

- **Grey box tri-state** at every level of the hierarchy — seder, masechta, perek, mishna. Empty, grey, checked.
- **Reorder → recompute.** Any change to the sequence discards whatever is sitting in backlog or today's schedule and replaces it with today's freshly computed estimate.
- **Free tick, and tick-to-here.** Browse the corpus and tick anything at any level, for out-of-schedule learning, for jumping ahead, and for backfill — which is the same screen used at a different moment.
- **Event dating.** Every event stores when it was *learnt*: dated, motzei-backdated, or before-tracking. Before-tracking counts fully toward the goal and the lifetime record, and never toward velocity.

- **Two types.** School year (year picker, rate, weeks) and ongoing (rate, bein hazmanim, weeks, optional end date).
- **Ordered, with one current position** — structurally the main track minus the plan and the deadline.
- **Drag in ground.** Any masechta, perek or mishna moves into a sub-track and out of the main track. Optional, incremental, phase two of onboarding a source.
- **Contribution maths.** Forward-only: past counted as fact, future as estimate, never both.
- **Recompute engine** — the one from stage 1, now with sub-track contribution in the numerator.
- **Home screen:** today's tasks on top exactly as they render now, sub-track position rows beneath.
- **Add next year** — clone the previous year's sub-track and shift its window.
- **Forecast guards:** the over-commit warning and gap detection.
- **Reports:** per-source velocity since he started, on or off target, PDF export.
- **Lifetime record:** times learnt per mishna, with dates, and a lifetime report.

STAGE 3

Shabbos

- **Shabbos-mode lock.** Capture closed from before candle-lighting until after tzeis,

driven by zmanim and one location setting.

Read-only, not invisible.

- **Motzei catch-up** card and reminder, generalising to yom tov at no extra cost.
- **Events dated to the day learnt**, so the record and the velocity report stay honest.

Throughout: the main track is structurally untouched. One masechta, one plan, one deadline, the same planner and the same sequencing. It only ever sees its remaining pool shrink and its daily target adjust.

WHAT THE EVENING WAS ACTUALLY ABOUT

Five things worth keeping.

1 The three “minor changes” were the big one.

The grey box was its representation layer. Reorder-recompute was its engine. Once both existed, all that remained of the hardest feature in the app was a **single column on the completion record** saying where the learning came from.

Two thirds of it was specced while being called small.

2 The most throwaway idea solved the hardest problem.

The session opened worried about duplication — school covering ground already learnt at home, and what to do about it. Hours later Daniel offered the lifetime record almost as an aside. There was never a duplication problem. There was **chazara**, and no schema could see that until the lifetime record named it.

3 **The hardest engineering problem was solved in the fourth message, and nobody noticed.**

Every country's school calendar differs; every school differs within a country; no calendar can be imported, inferred or configured. The coach spent three exchanges circling it. Daniel had already ended it with **“40 weeks per year”** — one integer that contains every yom tov, every term break and every country, and needs no maintenance, ever.

4 **The best design work was deletion.**

“Only the main track has a deadline.” “This should be impossible — don't change main track.” “One school per year, five others.” “Forget yom tov.”

The feature got better every single time something was removed from it.

5 **Shabbos changed genre.**

It came in as a scheduling exclusion — a checkbox next to yom tov and bein hazmanim. It left as the app's most distinctive feature: the thing that shows a frum family built it.

LEFT ON THE TABLE

Open, parked, and deliberately not built.

Still open

- ? **Fragmented history.** One school split across three year-sub-tracks shows up as three rows in the lifetime report rather than one “school”. Acceptable, or does it need grouping later?

Parked, not rejected

- **Letting the app learn the calendar.** If school posts nothing for three weeks each spring, the app has discovered Pesach without knowing the word. Revisit only if the manual estimate proves too inaccurate in practice.

? **The rebbe's non-active weeks.** Does the son learn with his rebbe through bein hazmanim and yom tov, or is that a different set of weeks again? The checkbox exists; the answer does not.

- **Confidence intervals** on the forecast, and **streak-safety** logic — cut under the simplicity mandate along with per-scope deadlines and any multi-option collision policy.
-
- **Warm and cold.** Because the log carries dates, the corpus could flag what was learnt recently and what was learnt once, years ago — a chazara prompt with no new storage. Offered, flagged optional, not adopted.
-
- **One number instead of two.** Stating a sub-track's estimate as a total for the year rather than a weekly rate would delete both the rate and the weeks field. Offered late, not taken.

WHOSE SESSION IT WAS

161 entries. Forty-four of them his.

One square per logged entry, in the order they happened. Wine is Daniel; the pale ones are the analyst. Hover any square to read it. Every structural decision in the design is a wine square.



■ Daniel – 44 (23 decisions · 11 ideas · 5 questions · 4 course corrections · the opening context)

Analyst – 116

Six of the analyst's squares are it withdrawing or correcting its own proposal — every one of them after Daniel pushed. The ratio is the wrong way to read this page: the analyst produced the volume, and Daniel produced the shape.

THE OPENING PROBLEM

“app assumes one curriculum -> one track -> one plan -> one goal. Reality is one corpus (all Mishnayos) fed by three sources”

Everything else on this page is downstream of it.

MINOR 1

“no partial-completion symbol. Completing one perek leaves the whole masechta reading unticked. Wants a GREY box (indeterminate) - empty / grey / checked.”

Became the representation layer of the entire feature and got reused in four places.

MINOR 2

“reordering the sequence changes 'next to learn'; any mishnayos sitting in backlog or today's schedule must be discarded and replaced with today's freshly-computed estimate.”

Became the recompute engine, and forced the derived-plan spine that makes out-of-order learning free.

MINOR 3 — AND THE CALENDAR PROBLEM, ENDED

“be able to enter expected weekly output from other programs - e.g. school 10 mishnayos/week, rebbe 5/week - plus active weeks per year (e.g. 40, since they break for holidays). Adjustable as his real pattern becomes clearer.”

One integer that contains every country, school, rebbe and yom tov. The model never needs to know *which* weeks.

SCOPE LOCK

“yom tov / bein hazmanim exist ONLY to estimate traction on the SUB-tracks (school, rebbe) ... Not a scheduling feature.”

Confined every calendar concept in the design to one number on one form.

SCOPE LOCK 2

“the whole daily-target computation is only relevant IF a deadline is set. No deadline, no required rate.”

Produced the build order: the zero-configuration version ships first and the estimates unlock behind a deadline.

NO SUB-GOALS

“nothing other than the main track ever gets a deadline. No 'finish Seder Moed before Pesach'.”

One deadline per curriculum, so the equation is solved once, globally. No competing deadlines, no nested progress maths.

THE GOVERNING PRINCIPLE

“keep it as simple as possible” — applied retroactively, pruning everything generated so far against it.

Cut inferred calendars, confidence intervals, streak-safety logic and multi-option collision policies in a single line.

THE LIFETIME RECORD

“every learning event is kept globally - tap a mishna and see how many times he has learnt it and when, across his whole life. Plus a lifetime report.”

Replaced the set with an event log, dissolved the duplication problem into chazara, and made the app outlive the goal.

SHABBOS

“the app must be NOT CLICKABLE on shabbos (and by extension yom tov) - capture is locked, no writing. After shabbos he is expected to tick what he learnt.”

The most distinctive thing in the design, and the one no generic tracker would ever have.

RESERVATION, SOLVED BY GESTURE

“you can DRAG any masechta, perek or individual mishna into a sub-track and it disappears from the main track.”

Delivered his opening carve-out requirement with no extra field on the form — and revived the double-count landmine, which is what produced the `min()` in the equation.

TWO TYPES

“there ARE two sub-track types. A SCHOOL sub-track runs September to July and lasts exactly one year. Not an arbitrary window.”

Overruled the analyst. The type is what lets the app fill the dates, offer “add next year”, drop the exclusions and refuse nonsense windows.

ONE POSITION PER TRACK

“EACH TRACK HAS EXACTLY ONE CURRENT POSITION: a sub-track holding several masechtos needs an ORDER, like the main track. Only one thing is being learnt at a time per track.”

The cleanest piece of architecture in the session, and better than the analyst's. Every track is an ordered sequence with one position; only the main track adds a plan and a deadline.

THE REVERSAL

“two masechtos on the MAIN track must be IMPOSSIBLE.”

Left the planner, task generation and sequencing entirely untouched, and deleted an open question in the same breath.

LIMITS

“one school sub-track per academic year, from now until the goal; and up to FIVE other (ongoing) sub-tracks.”

Turned validation into a picker: the impossible choice is simply not offered.

THE LAST TRIM

“drop yom tov as a checkbox entirely; keep BEIN HAZMANIM only ... And 'average across the year' is confusing wording.”

One checkbox left, and the word “average” deleted in favour of two visible numbers that multiply.

הדרן עלך

Six sedarim, sixty-three masechtos, about four thousand one hundred and ninety-two mishnayos, and a boy learning them at home, with his rebbe, and at school — all of it landing in one place.

The record does not end at the siyum. The bar mitzvah is one milestone in a record that keeps going.

Brainstorming session, 24 August 2026 · Learning Tracker · every word on this page derives from the session log of 161 entries.